1. Blockchain-Based Decentralized Smart ride pool system for IIEST, Shibpur.

Here blockchain will handle several parts –

- ride agreements.
- Passengers commitment.
- Payment.
- Penlaty if not come or Late come.
- Driver's payment system.

Step1.

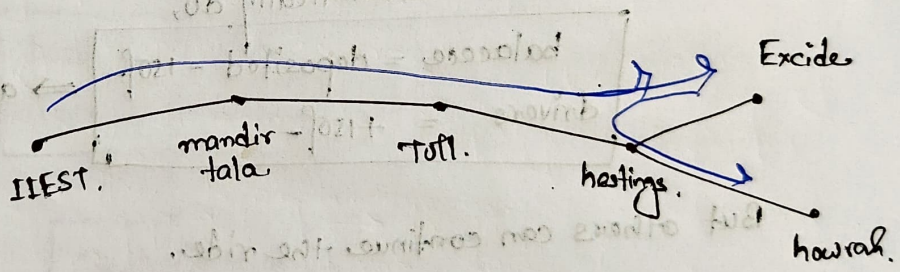
First of all. students request the ride through ride pooling app by entering the details.

Pickup : IIEST. gate no. 1.

Drop : Howrah station route.

Step2

↳ Now there should be an algorithm that 4 passengers if the route is same (or almost) accept the ride, → ride confirmed.



Step3.

↳ Now blockchain part is implemented.

Smart contract generated.

Ride id ().
Driver id ().
Passengers ids ()
Pickup location.
Drop location.
time
Penalty rule → if not come or late.
Fare detail

it is stored in the blockchain. (cannot be edited once created)

Step 4.

↳ Payment Locking.

Each students should have their virtual wallet. They should Load fair into smart contract wallet.

let's assum. 150/- per person, so, total 600/- is now locked in the contract.

Step 5:

Pickup.

↳ match passengers GPS with car's GPS + QR or OTP

Step 6.

If passengers are late or does not come

Smart contract automatically do,

balance = deposited - 150/-	⇒ automatically.
drivers = +150/-	

But others can continue the ride.

Step 7.

Now some of them drops at diff. Location,

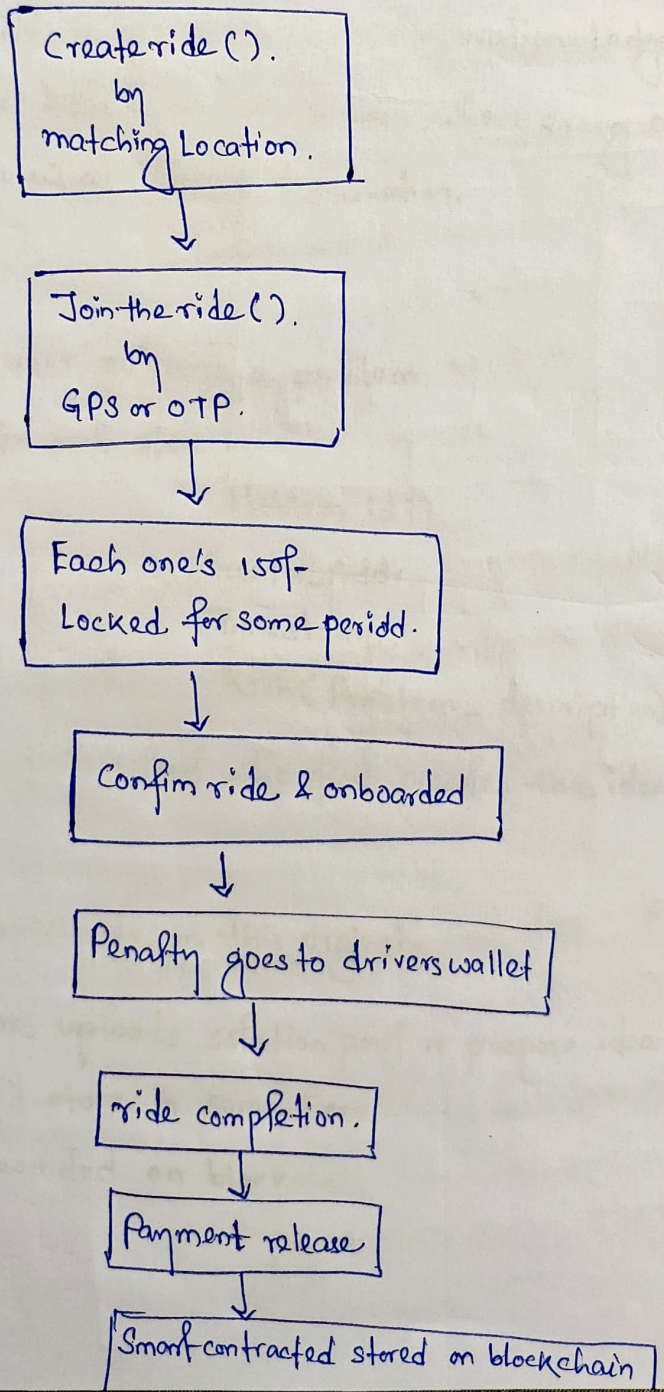
Step8.

Drop-off.

Now Smart contract releases the payment.

→ payment goes to driver's virtual wallet.

Now smart contracted is stored on the blockchain.

↓

Trustworthy, transperency, automatic penalty.